WORKED SOLUTION

STEP I 1989, Question 4

Pure • Euclidean geometry

STEP I 1989, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q4

This question has a full guided treatment in *Beyond STEP*, Section 9.2, *Ramsey's Theorem in Miniature*: https://beyondstep.keithmcnulty.org/ch09/ramsey.html

The key idea

Forget the geometry: the conditions on the points only guarantee that any three of them form a genuine triangle, so this is purely a question about colouring the 15 edges joining six points. Following the hint, look at one point. It has five edges and only two colours, so by the pigeonhole principle at least three of those edges share a colour. Then the edges among the three far ends either complete a one-coloured triangle with the first point or form one on their own.

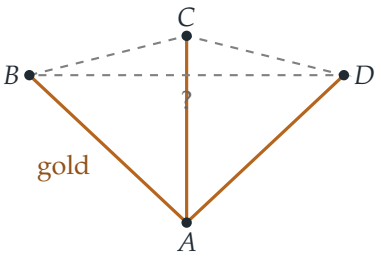
Solution

Six points

Since no three of the points are collinear, any three of them are the vertices of a genuine triangle, and its three sides are among the segments we have drawn. (The condition that no four are coplanar plays no part in the argument.)

Consider the five segments AB, AC, AD, AE, AF that come out of A. Each is gold or silver. If at most two were gold and at most two were silver, there would be at most four segments in total, which is false. So at least three of them have the same colour. By relabelling the points $B, \ldots, F$, and swapping the names of the two colours if necessary, we may assume that

$$AB,\ AC,\ AD \text{ are all gold.}$$



Now look at the three segments BC, CD and DB joining the far ends.

Ask yourself Why must one of the two cases below give a one-coloured triangle, whatever colours BC, CD and DB have?

There are only two possibilities.

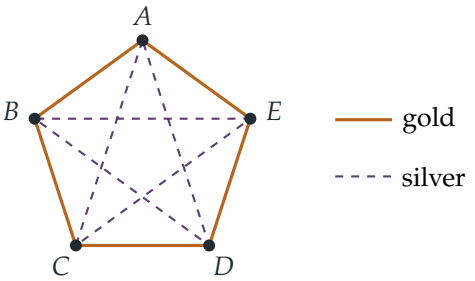
- *At least one of BC, CD, DB is gold.* Say it is BC (the other cases are the same with the letters changed). Then AB, AC and BC are all gold, so triangle ABC has edges of one colour.
- *None of BC, CD, DB is gold.* Then all three are silver, and triangle BCD has edges of one colour.

Either way there is a triangle whose edges are all the same colour, which is what we had to prove.

Five points

With five points, each point has only four edges, which can be split two and two, so the argument breaks down. To show the result really is false, we need one colouring with no one-coloured triangle.

Place the points A, B, C, D, E round a pentagon. Colour the five sides AB, BC, CD, DE, EA gold, and the five diagonals AC, BD, CE, DA, EB (the pentagram) silver.



Any triangle uses three of the five points. Among any three vertices of a pentagon, at least two are not next to each other (three mutually adjacent vertices would need a triangle inside the 5-cycle), so every triangle has a silver edge. Also, among any three

vertices at least two *are* next to each other: if no two of the three were adjacent, each gap between consecutive chosen vertices round the pentagon would contain at least one unchosen vertex, needing $3 + 3 = 6$ points. So every triangle also has a gold edge. Hence no triangle is all one colour.

Ask yourself A regular pentagon is flat, so four of its vertices are coplanar. Doesn't that break the rule about general positions?

It would, if the points were literally on a plane, but the colouring depends only on which pairs are joined, not on where the points are. Lift the vertices slightly off the plane by different amounts (for example, raise the kth vertex to height $0.3k^2$ for $k = 0, 1, 2, 3, 4$). Then no three are collinear and no four are coplanar, and seen from above the picture is still the pentagon and pentagram drawn here, with the same colouring and no one-coloured triangle.

Checks

There are only $2^{15} = 32768$ ways to colour the edges for six points, so a computer can try them all: every one has a one-coloured triangle, and in fact always at least two. For five points, checking all $\binom{5}{3} = 10$ triangles of the pentagon colouring confirms that each has both colours. A determinant check confirms that the lifted pentagon above has no four points coplanar.

Common slips

- Saying "by symmetry" without saying what is being assumed: be explicit that you choose three edges from A of the majority colour and call that colour gold.
- Only proving that A has three edges of one colour and stopping there; the second step, looking at BC, CD, DB, is the heart of the proof.
- For five points, giving a sketch without checking every triangle; a colouring such as "a star from A gold, everything else silver" fails.

Beyond STEP

This is the first case of Ramsey theory: the *Ramsey number* $R(3,3)$ is the smallest n such that every two-colouring of the edges of the complete graph K_n contains a one-coloured triangle, and this question shows that $R(3,3) = 6$. Ramsey's theorem says such a number exists for any sizes of one-coloured complete subgraph, but they are fiendishly hard to compute: even $R(5,5)$ is unknown. A lovely introduction to graph theory, including Ramsey's theorem, is Wilson, *Introduction to Graph Theory*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.